TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 1

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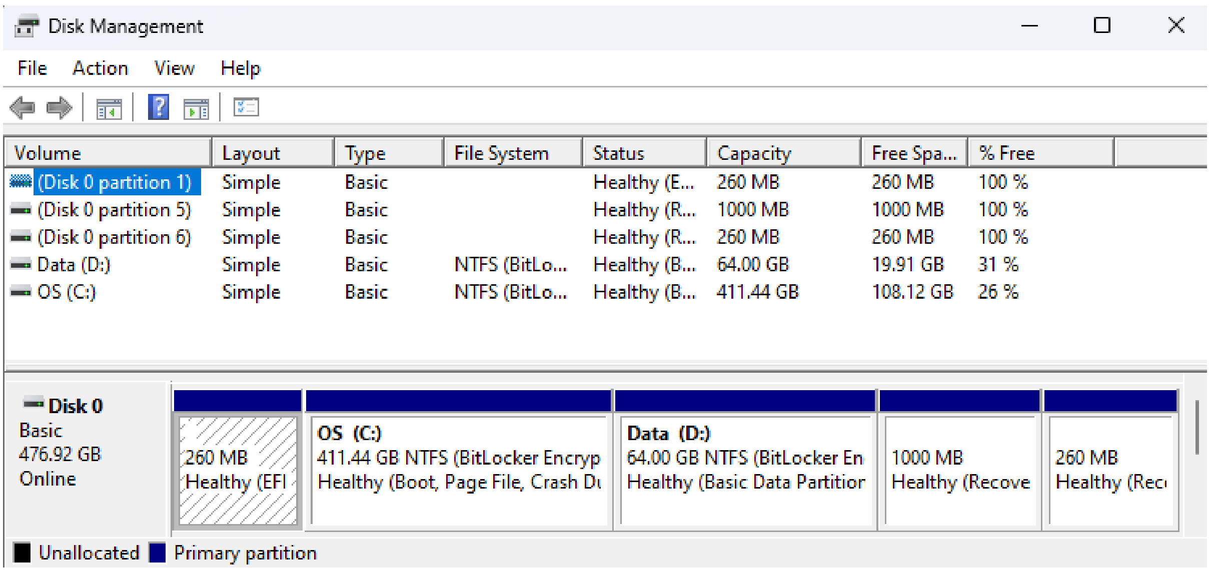
Section	SAM1, 1A

1. You need to install Linux server in the Lab PC using VirtualBox. Identify the **minimum** requirement on RAM and disk space for THREE (3) most popular Linux server namely Fedora, OpenSuSE and Ubuntu. Compare the requirements of OpenBSD, FreeBSD and NetBSD.

Distro	Fedora	OpenSuSE	Ubuntu	OpenBSD	FreeBSD	NetBSD
RAM	2GB	2GB	4GB	256MB	512MB	64MB
HDD	8.95GB	7.59GB	6.06GB	670.05MB	1.09GB	622.42MB

2. Identify the CPU type and speed, RAM size and hard disk size of your PC/notebook. Use and show command/program in Windows or Linux to be used during the identification process.



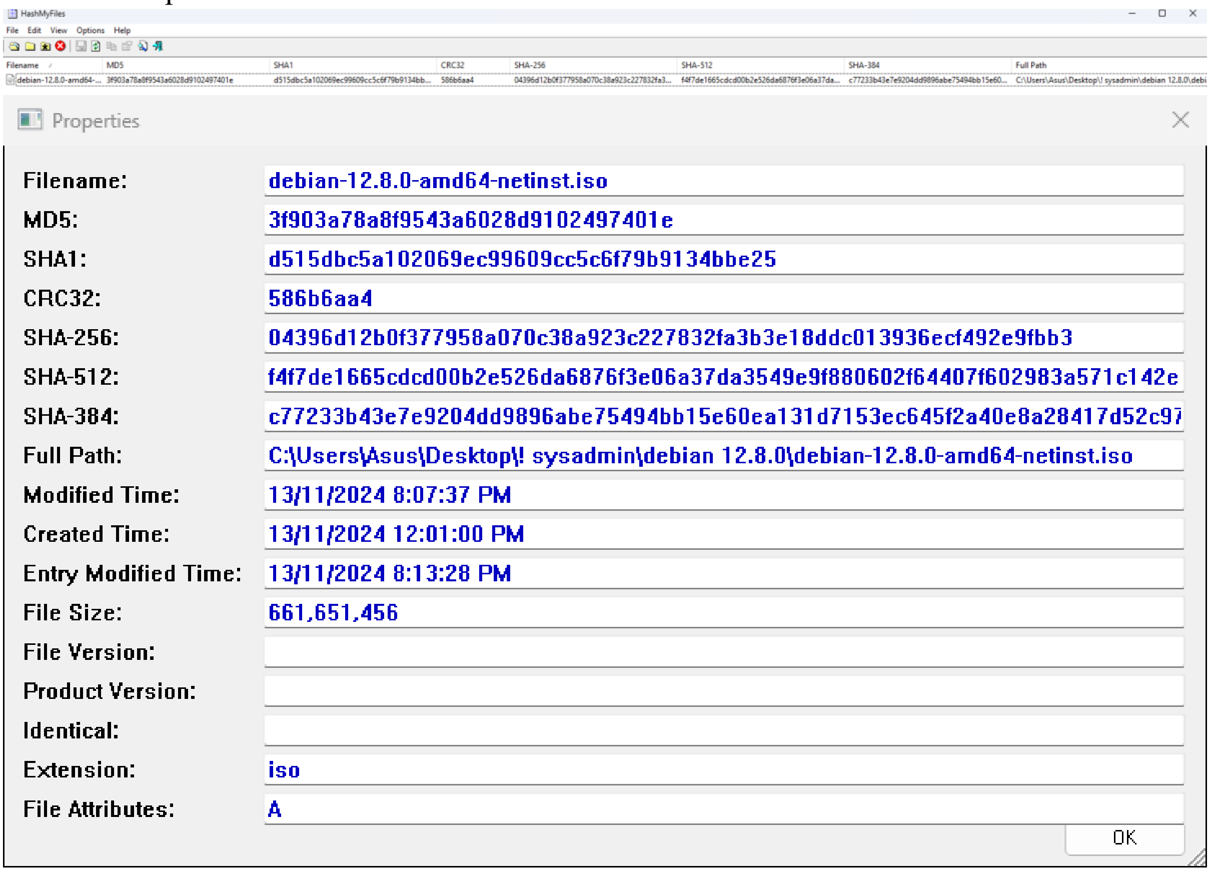
3. Most Linux distro are freely available and distributed in the format of ISO files. Normally it comes with MD5 or SHA sum file. Explain what ISO file is, MD5 and SHA sum.

ISO is identical copy or image of data found on an optical disc such as CD and DVD.

MD5 is message digest 5

SHA is secure hash algorithm or SHA sum.

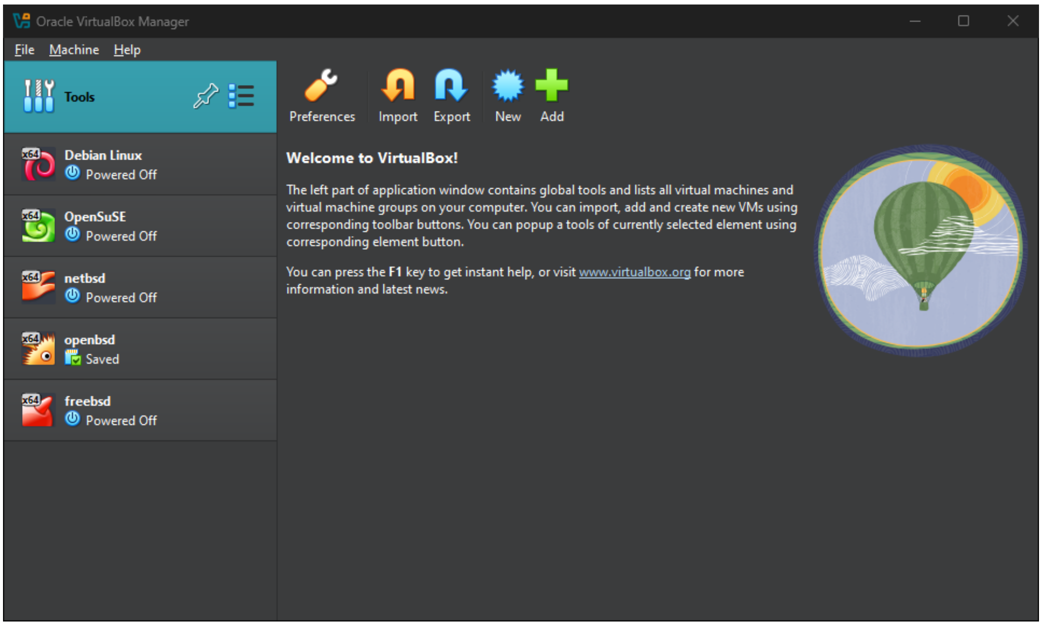
4. Download Debian ISO file and compute and verify their sum using certutil, HashMyFiles http://www.nirsoft.net/utils/hash_my_files.html, md5deep, mdsummer or any other program that can compute hash for a file.



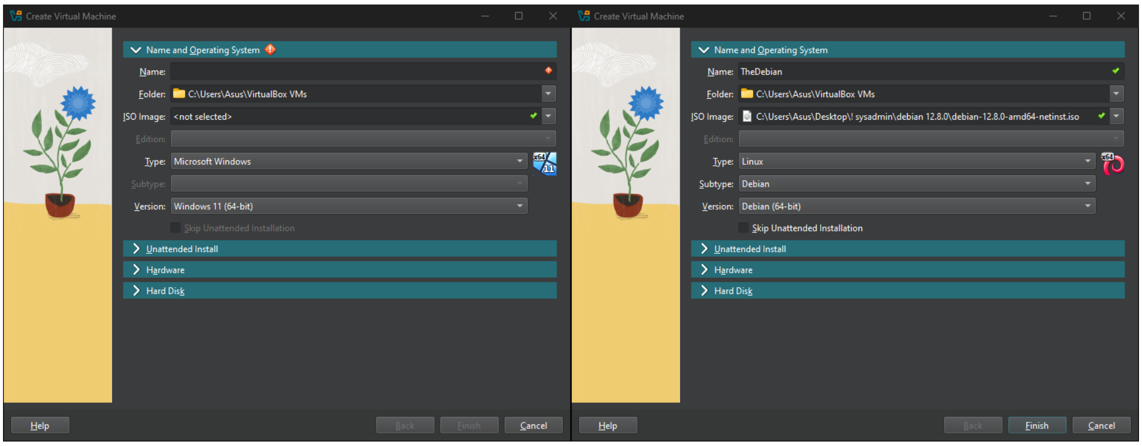
5. Install Debian Linux in your Lab PC using VirtualBox. In VM setup, set 2GB RAM and 10GB disk size. Capture the installation process.

Redo – need ss every stages and screen.

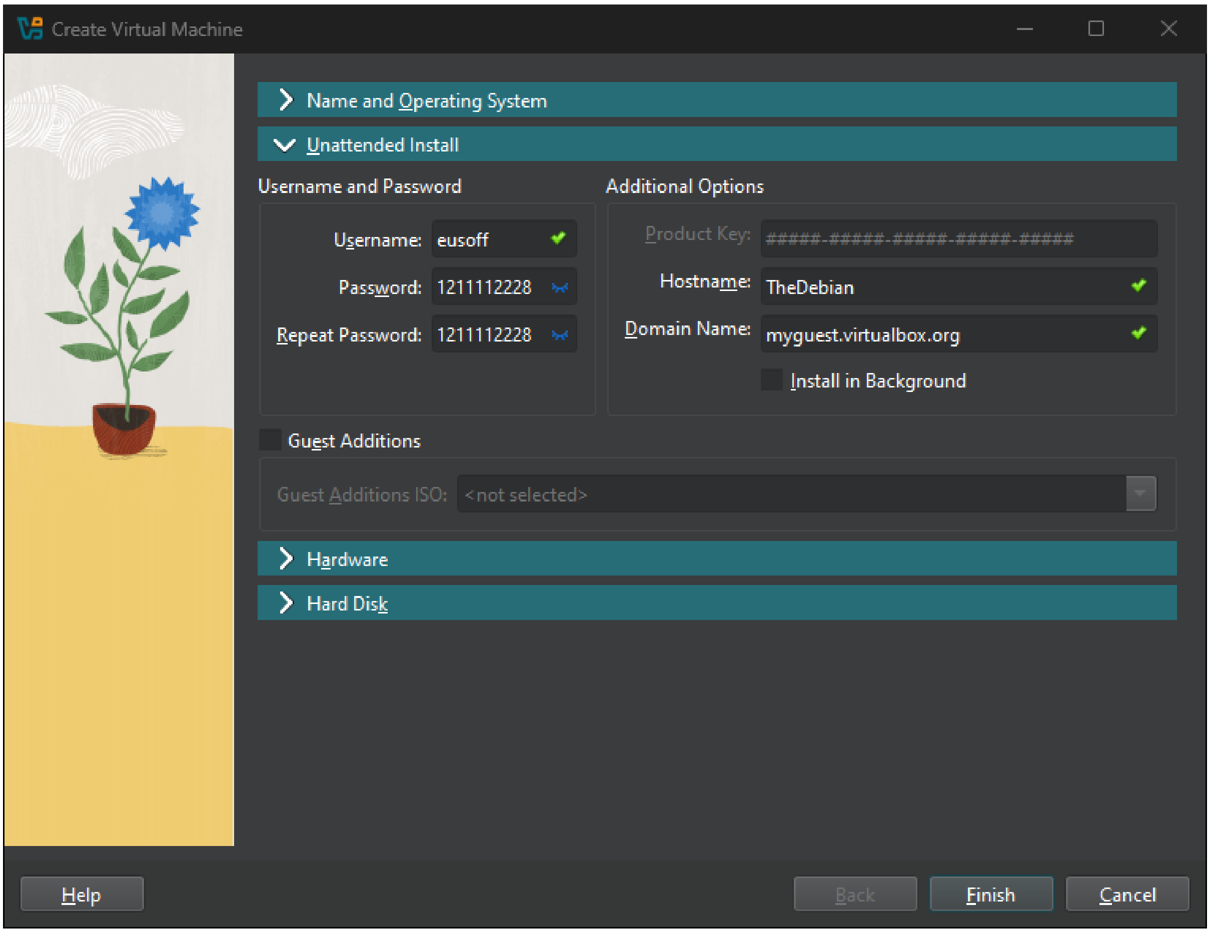
Click on New to create a new Debian Linux VM.



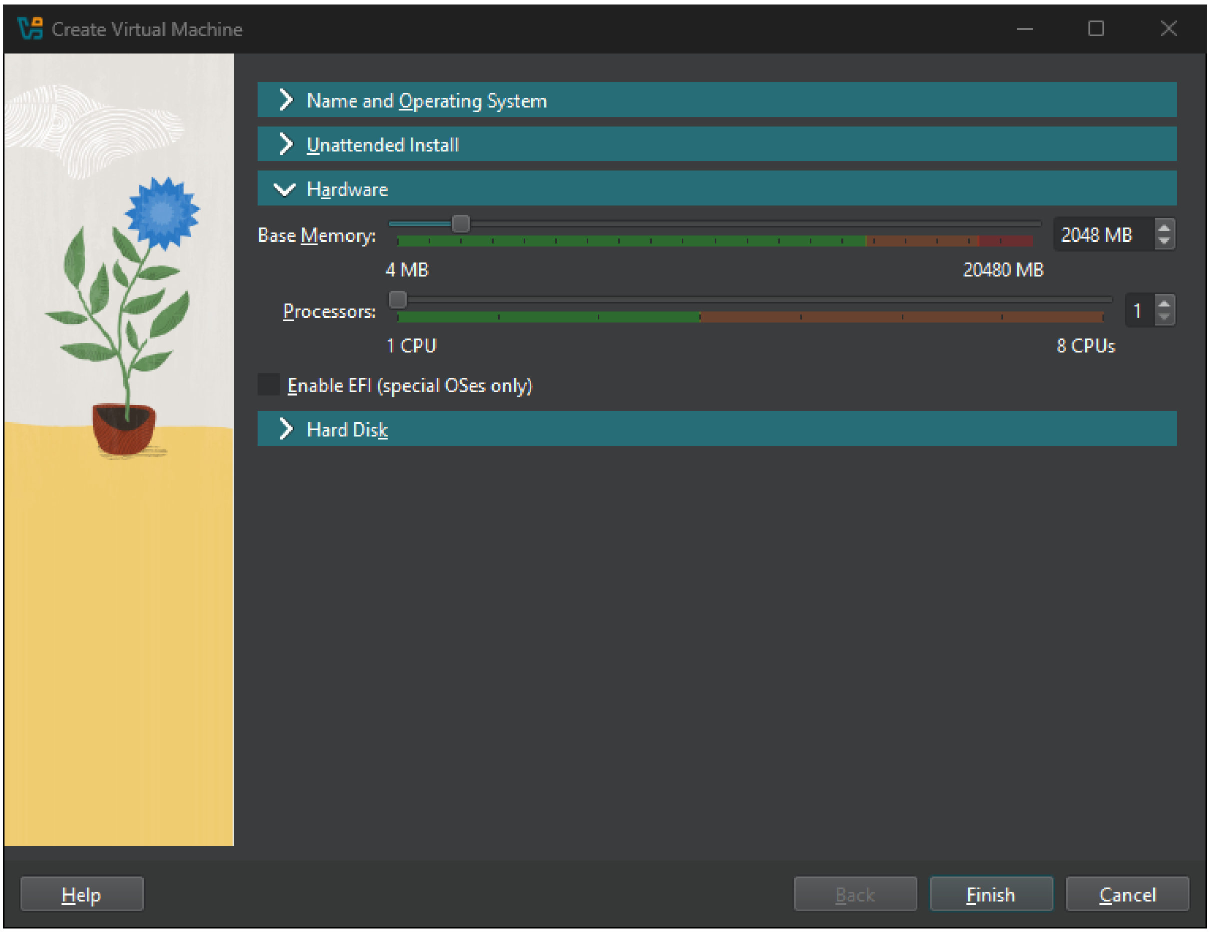
Under "Name and Operating System", name the VM and select the Debian .iso file. Type, Subtype, and Version will automatically be selected after naming the VM.



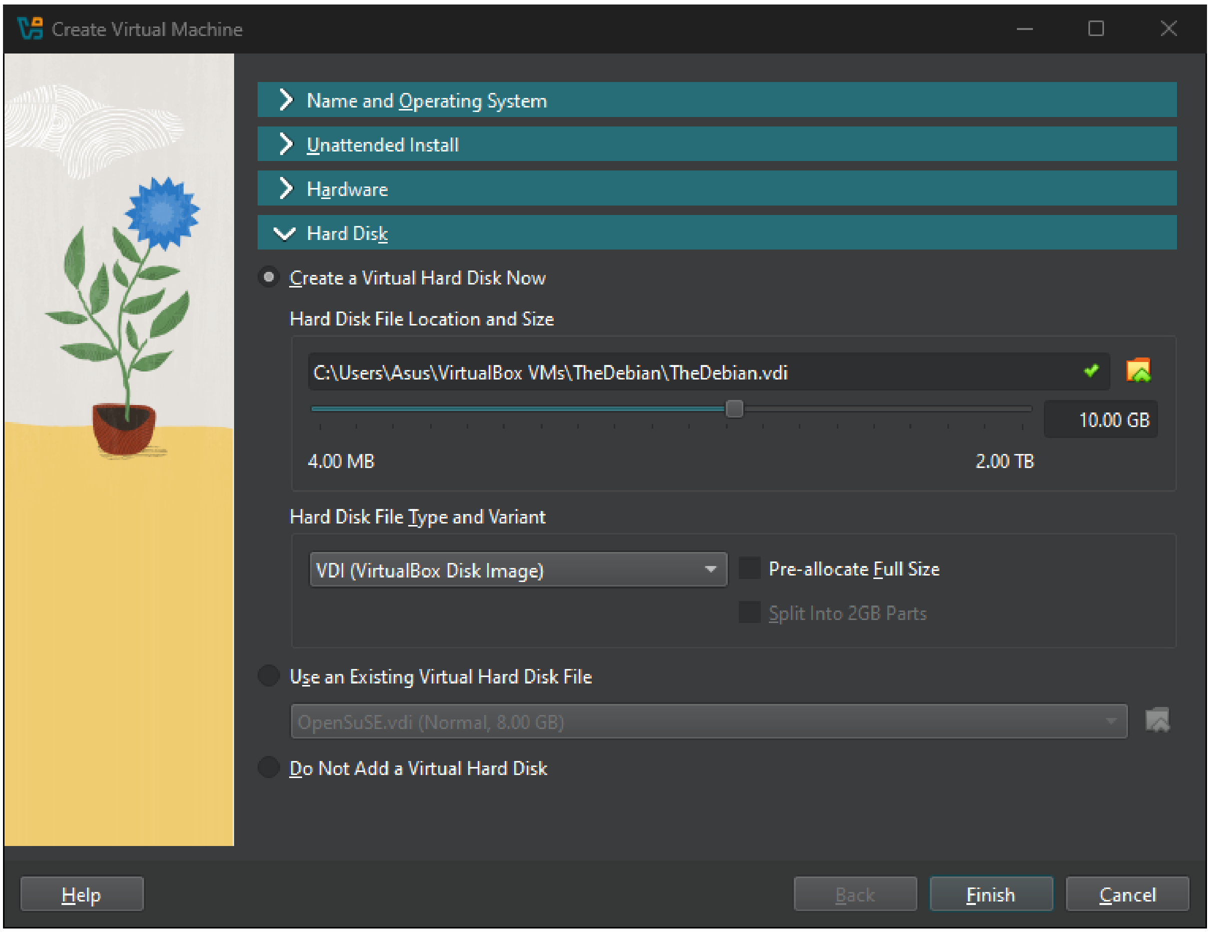
Under “Unattended Install”, you can rename the username and password. In this example, username is “eusoff”, and password is “1211112228”.



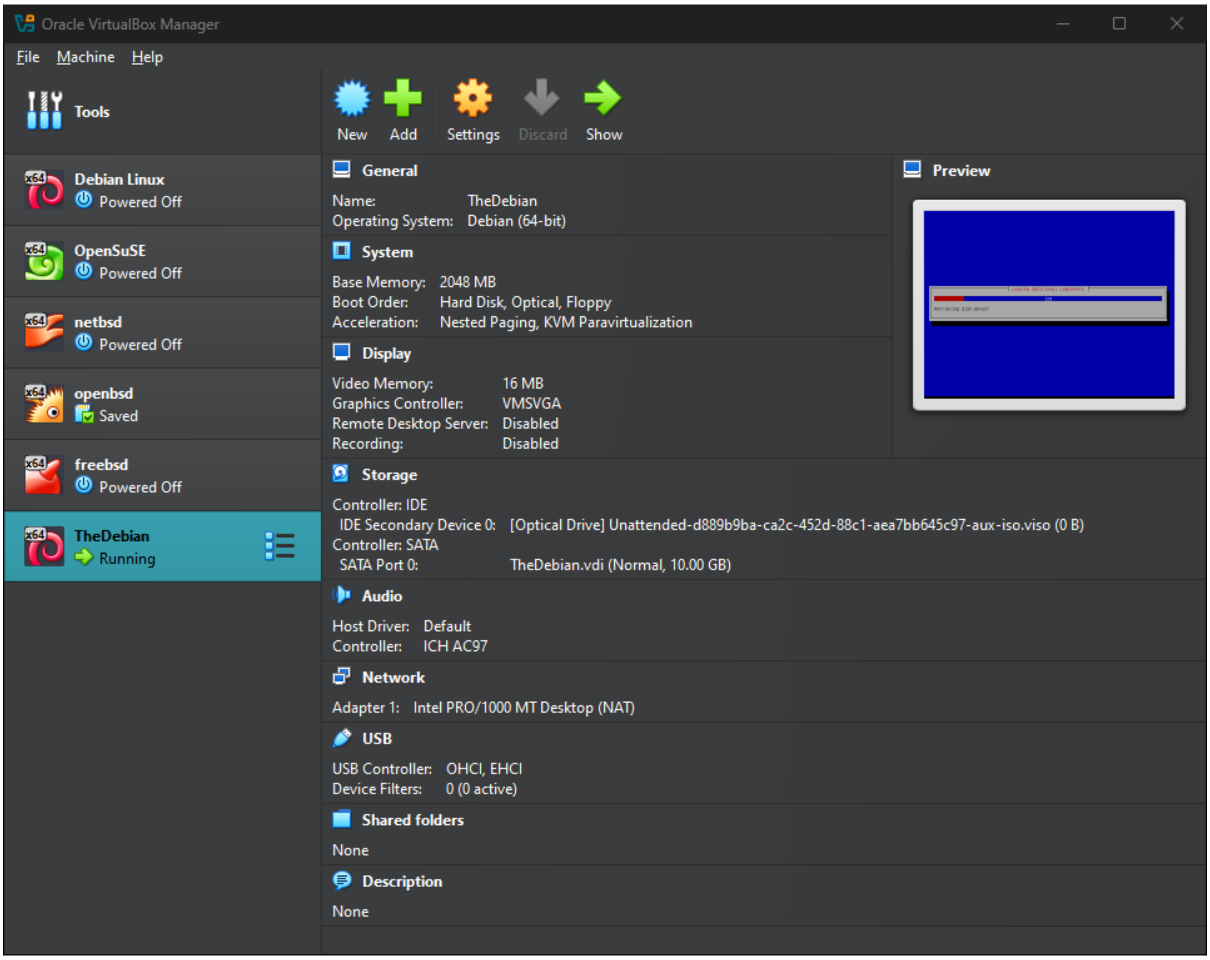
Under “Hardware”, we can set the base memory, which is RAM, and how much CPU processors count. In this guide, I will select 2GB RAM and 1 CPU.



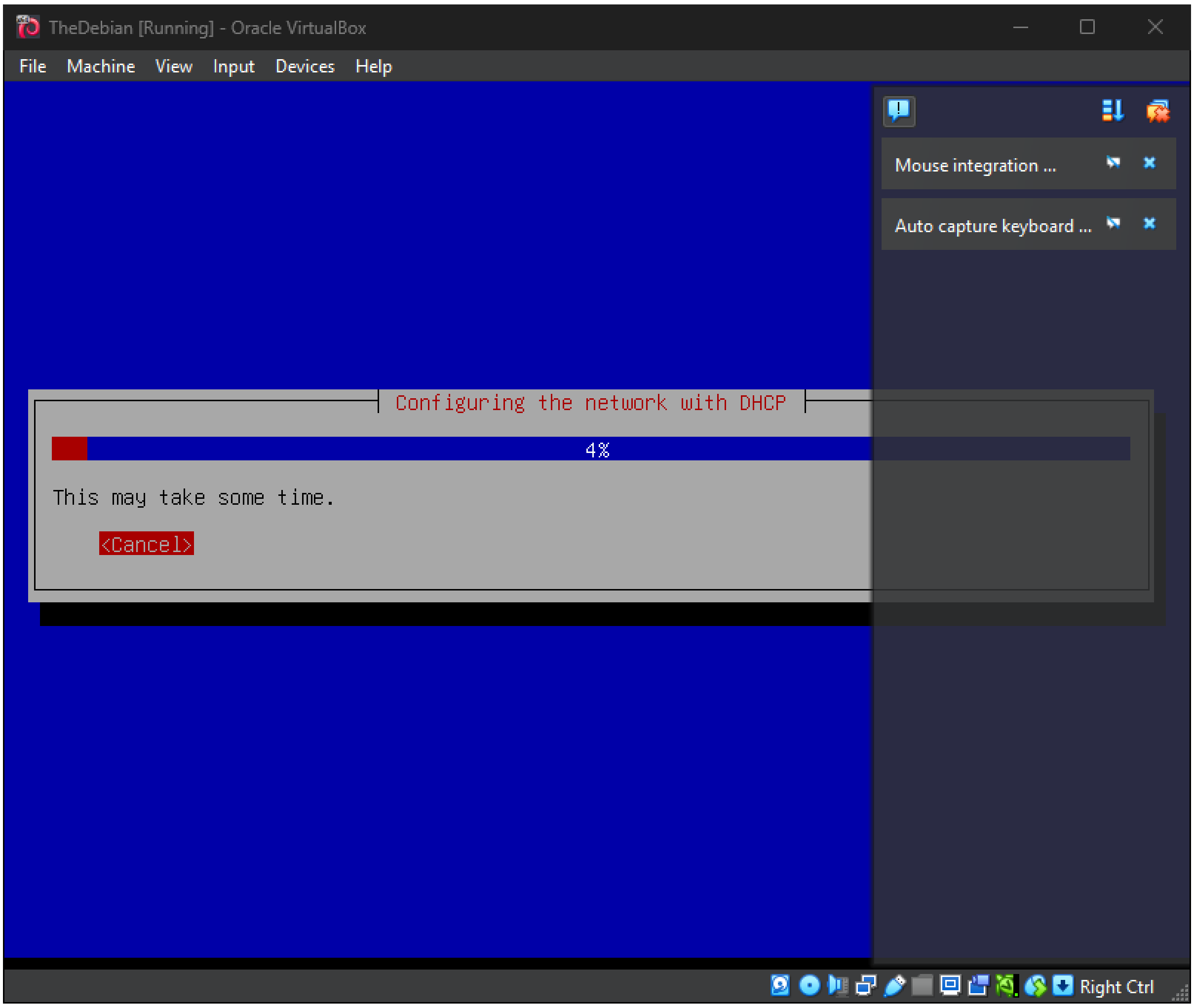
I will select 10GB of storage for this VM.



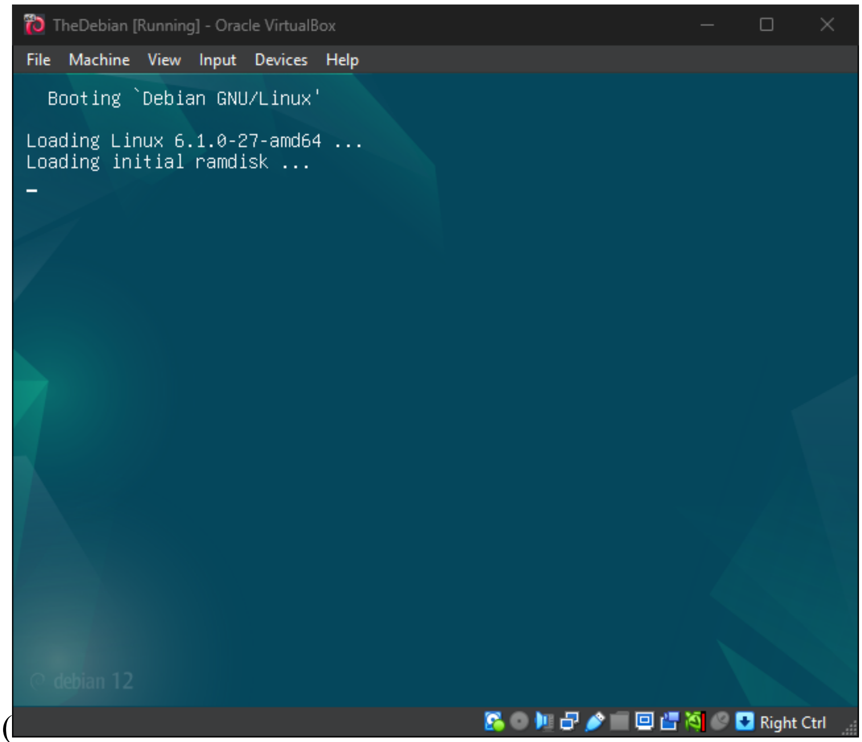
And then, click “Finish”. VM will ran automatically.



Let the VM run and install. May take some time.



Once done installing, it will reboot and allow us to choose which bootloader to boot from.



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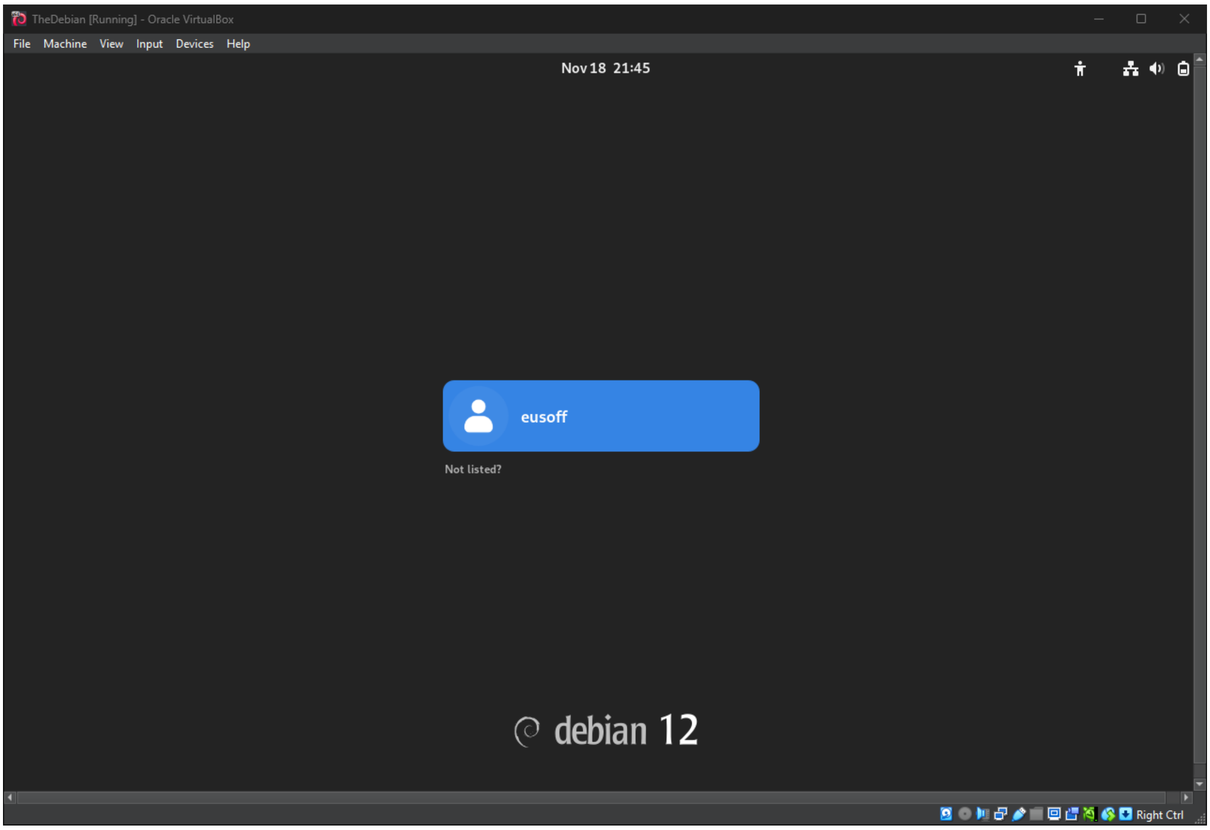
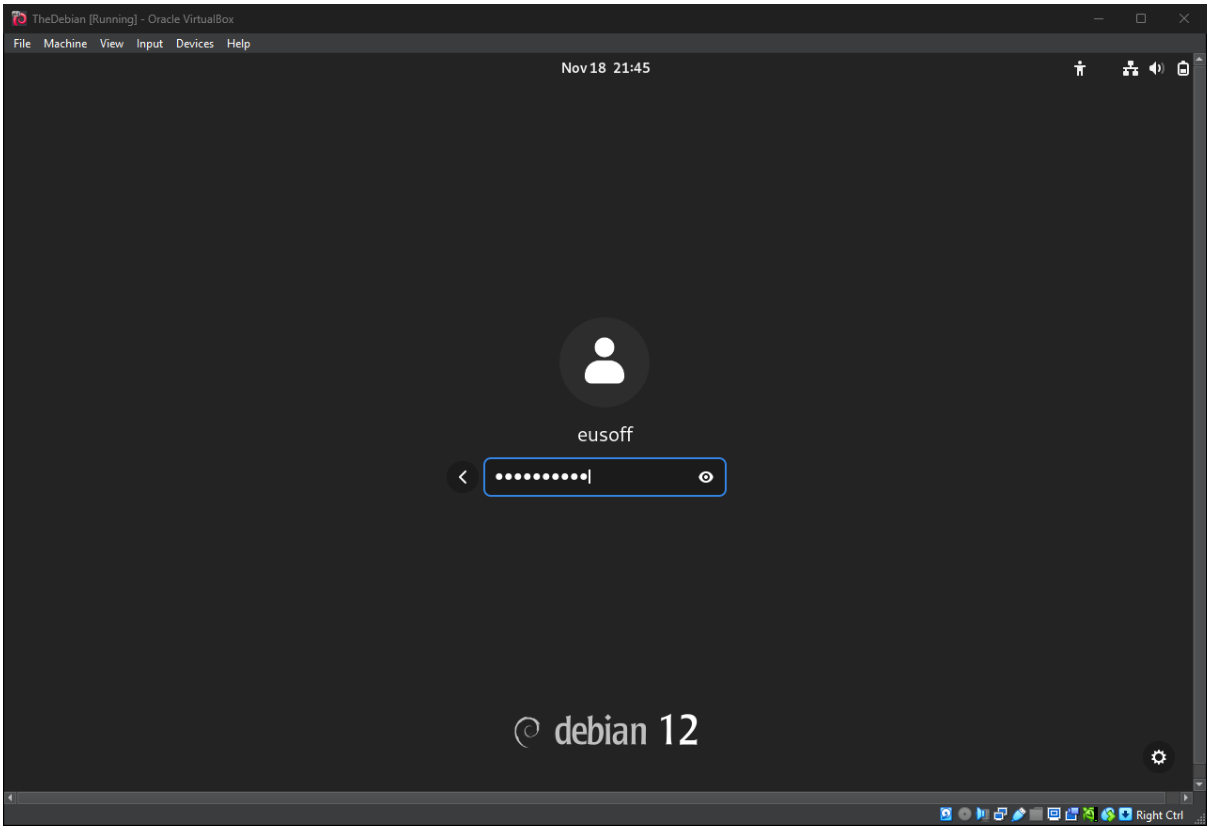
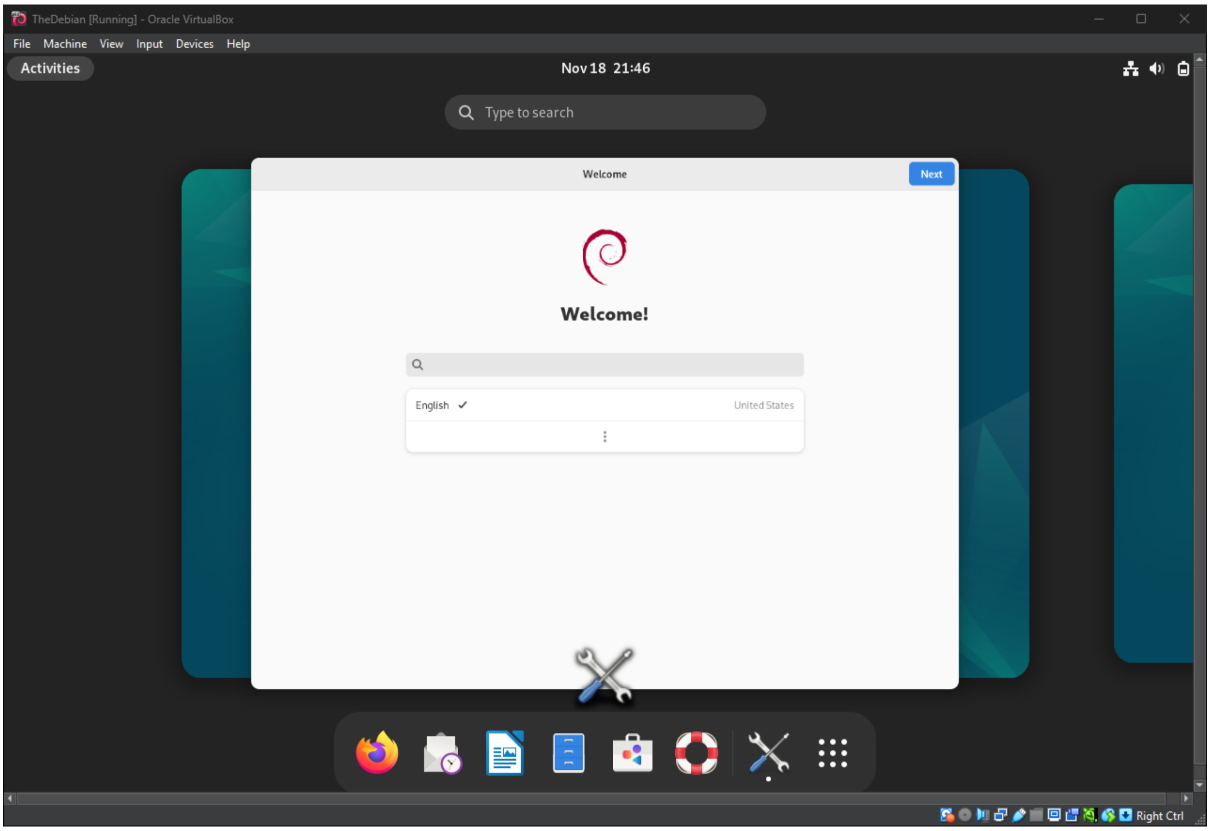
And lastly, we are able to login.



Enter the password for our username and we will be able to log in.



Once this have popped up, we are free to use Debian Linux.



Open setting, and click on About. It will show the specifications of the VM.